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Review of AI methods in precision agriculture

El Hou Meryem | Zahidi Yassine | Rabbah Nabila | Eldin Mohamed | Medromi Hicham | Touati Abdelwahed

Agricultural and Food Sciences, Computer Science · E3S Web of Conferences · 2025

TLDR The **review** reinforces the idea that AI technologies can boost the resiliency, productivity, and sustainability of agriculture and believes that AI specialists and agricultural scientists collaborating on affordable, reliable, and field-ready innovations would have the greatest impact on stimulating widespread adoption. Expand

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AI-Driven Disaster Prediction and Early Warning Systems: A Systematic Literature Review

K. Luxshi Environmental Science, Computer Science ·

International Journal of Research in Computing · 1 July 2024

TLDR This research explores how AI technology connects with the Internet of Things (IoT) and remote sensing systems for real-time disaster monitoring and identifies ways to enhance AI-based disaster resilience strategies. Expand

1 Publisher Save Cite

Data-driven methods in risk assessment of public-private partnership projects

Fatemeh Khalilnezhady | Pauline Teo | B. Trigunarsyah | Nader Naderpajouh

Engineering, Computer Science · Engineering Construction and Architectural… · 13 October 2025

TLDR The research fills a critical gap in the literature by highlighting the need for interpretable AI models to improve the transparency and practicality of AI applications in PPP risk **management**. Expand

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Investigation of communication systems powered by AI for autonomous transportation

Shivangi Bansal | Shivangi Gupta | +4 authors | Ranganathaswamy Madihalli Kenchappa

Engineering, Computer Science · Multidisciplinary Reviews · 1 December 2025

TLDR This **review** investigates how AI approaches, notably **deep learning** (DL) and **machine learning** (ML), improve autonomous vehicle communication, traffic **management**, and security, and compiles research on **deep learning** models such as Convolutional Neural Networks, Long Short-Term Memory (LSTM) networks, and Reinforcement **Learning** (RL), as well as **machine learning techniques** such as Random **Forest**s (RF) and decision trees (DT). Expand

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Integrating AI in Structural Health Monitoring (SHM): A systematic review on advances, challenges, and future directions

Mengesha Azanaw Girmay Engineering, Computer Science ·

i-manager's Journal on Structural Engineering · 2024

TLDR The continuous maturation of AI is sure to continue its application in SHM and thereby help develop a host of safe, resilient, and sustainable infrastructure systems as one of the key future developments in structural engineering. Expand

11 Publisher Save Cite

AI Driven Innovation for Boosting Performance and efficiency in Mobile and Wireless Networks

Arjola Biti · Olimpjon Shurdi · Luan Ruçi · Computer Science, Engineering ·

2024 5th International Conference on… · 20 November 2024

TLDR The study explores how **artificial intelligence** algorithms can transform network optimization and **management**, to achieve superior performance and reliability, by automating optimization processes, minimizing operational costs, and flexibly adjusting to evolving network conditions and user requirements. Expand

3 · IEEE · Save · Cite

Review of Predictive Modeling Techniques in Financial Services: Applying AI to Forecast Market Trends and Business Success

Ayodeji Ajuwon · Ademola Adewuyi · Omoniyi Onifade · Tolulope Joyce Oladuji

Computer Science, Business · International Journal of **Management** and… · 2022

TLDR This **review** synthesizes current predictive modeling **techniques** applied within the financial sector, emphasizing the integration of AI methodologies for forecasting market movements and assessing business success. Expand

3 · Publisher · Save · Cite

A Systematic Analysis on Crime Detection and Prevention Through ML Communication Techniques

Saroj Kushwah · U. Jaiswal · Jaishree Jain · Eshank Jain · Bharat Singh · Parth M Garg

Computer Science, Law · 2025 **Artificial Intelligence** and Smart… · 21 August 2025

TLDR This abstract aggregates results from various geographical contexts, hence proving the universal value of ML-based crime prediction systems in developing proactive crime **management** strategies. Expand

IEEE · Save · Cite

A Review of AI-Driven Engineering Modelling and Optimization: Methodologies, Applications and Future Directions

Jian-Ping Li · N. Polovina · Savas Konur · Engineering, Computer Science · Algorithms ·

23 January 2026

TLDR A critical assessment of the state of the art of AI-driven optimization methodologies is provided and promising research directions are identified, including physics-informed neural networks, digital twins, and human–AI collaborative optimization frameworks are identified. Expand

1 · Publisher · Save · Cite

Diagnostic Applications of AI in Sports: A Comprehensive Review of Injury Risk Prediction Methods

C. Musat · C. Mereuță · +7 authors · L. Nechita · Computer Science, Medicine · Diagnostics ·

1 November 2024

TLDR This **review** highlights AI's ability to analyze complex datasets, detect patterns, and generate predictive insights that enhance injury prevention strategies that contribute to enhanced athlete safety, optimized performance, and reduced human error in medical decisions. Expand

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